Data Mining Project: Analyzing Reader Patterns and Genre Associations in the Goodreads Dataset*

* CS 4412: Data Mining

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***Abstract*—This proposal outlines a data mining project focused on the 2017 Goodreads dataset. This study aims to discover relationships between book genres, user behaviors, and how readers structure their reviews. We will be applying different techniques such as Association Rule Mining, K-Means Clustering, and Latent Dirichlet Allocation. This proposal seeks to understand how readers associate different genres in an attempt to define different segments of the reading community.**

***Index Terms*—Data Mining, Goodreads, Association Rules, Clustering, Text Mining, Pattern Discovery**

I. Introduction

The digital age has transformed reading from a personal, solo activity into a social activity. Various platforms like Goodreads and Fable provide a massive database of user-generated data through ratings, reviews, and custom shelves. While many similar projects exist to predict user's rating there is more value in understanding how users engage with literature. This project focuses on the UCSD Goodreads dataset to find these patterns.

II. Description

This project will utilize the Goodreads Book Datasets (2017) curated by Julian McAuley at UCSD, Rishabh Misra and Ndapap Nakashole

A. A. Source and Scale

- **Source:** https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html
- **Size:** The full dataset contains metadata for 2.36 million books and 15.7 million detailed review texts. For this project, a subset (e.g., the "Young Adult" or "History" category) will be utilized to maintain computational feasibility while ensuring a sample size of over 100,000 records.
- **Format:** Data is provided in the CSV and json format, requiring significant preprocessing and flattening.

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B. Key Features and Schema

The dataset is split into two primary components:

1) **Metadata (*goodreads_books.json*):** Contains `book_id`, `title`, `authors`, `average_rating`, and importantly, `popular_shelves` (user-defined tags).
2) **Reviews (*goodreads_reviews.json*):** Contains `user_id`, `book_id`, `rating`, `review_text`, and `n_votes`.
3) **User Interactions (*goodreads_interactions.csv*):** Contains `user_id`, `book_id`, `is_read`, `rating`, and `is_reviewed`.

III. C. Data Quality and Preprocessing

Known issues include many user-defined shelves that share common tags like "to-read" and varying review lengths. This is most likely caused from Goodread's ability to easily add books to user's "to-read" shelve. Preprocessing will involve tokenization, stop-word removal for text mining and transforming the nested JSON shelves into a transaction-style for association mining.

IV. Discovery Questions

This project moves beyond prediction models to investigate the following questions the team had:

- What are the hidden associations between specific book genres and user-defined "shelves"? (Do readers of "Romantasy" (Romance Fantasy) also show high interest in other genres like fantasy?
- Can we identify distinct Reader Personas based on interaction metrics? By clustering users based on their average rating variance, review frequency, and review length our goal is to discover if it is possible to classify readers into different groups?

V. Planned Techniques

The analysis will follow a multistage data mining pipeline as visualized in Figure 1.

Analysis Pipeline Diagram

1. Data Ingestion: Books & Reviews JSON Parsing

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2. Preprocessing: Admin Tag Cleaning & Language Filtering

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3. Mining Modules & Evaluation:

→ *Shelf Tags* ⇒ **FP-Growth** ⇒ Genre Associations (DQ1)
→ *User Metrics* ⇒ **K-Means** ⇒ Reader Personas (DQ2)
→ *Review Text* ⇒ **LDA** ⇒ Helpfulness Patterns (DQ3)

Fig. 1. Planned Data Mining Workflow mapping specific data features to their respective algorithms and discovery questions.

A. Association Rule Mining (FP-Growth)

We will treat each user's shelves as a transaction. Using the FP-Growth Algorithm to discover rules with high lift and confidence to identify non-obvious genre relationships. Our goal is to move beyond the standard "Fantasy" label and discover more sub-genre patterns.

B. Clustering (K-Means)

To answer the second discovery question, we will apply K-Means clustering on standarized numerical features derived the user dataset. We will be using the "Elbow Method" and "Silhoutette Score" to determine the optimal number of reader segments.

C. Text Mining (LDA - Topic Modeling)

Using the Latent Dirichlet Allocation (LDA) algorithm, we will perform topic modeling on the `review_text`. This will allow us to discover the primary topics of discussion within different genres without pre-defining what those topics are.

VI. Preliminary Timeline

The project will proceed according to the following milestones

TABLE I
PROJECT MILESTONE TIMELINE

Milestone	Period	Key Activities
M2	Weeks 8	Applying KDD process and Association Rule Mining
M3	Weeks 11	Clustering Analyses, Classification Techniques, detecting Anomalies, and Communicating results
M4	Weeks 14	Text Mining (LDA), Final Report

A. Anticipated Challenges

- Our primary challenge is lack of experience in analyzing data sets.
- Another challenge is the volume of data. The Goodreads dataset is comprised of multiple csv and jsons files that is multiple gigabytes compressed. Handling this in local memory will require using pandas chunking or focusing on smaller genre subsets. Text data can also be "noisy" and will require filtering to produce meaningful topics.
- Due to Goodreads' global reach our team expects to encounter non-English reviews, books, and shelf titles. Our team will have to strip any non-English text out of the datasets in order to prevent any of our proposed tests from failing.

VII. Milestone 2

This section details the work done for Milestone 2 of this data mining project which focused on the initial implementation of an exploratory discovery analysis. Our primary goal for this milestone was to address the first of our three discovery questions through data cleaning, analysis, and visualization Our team analyzes the most frequent tags of users' bookshelves to gain a first look at how users categorize books on the site.

A. Preprocessing and Data Transformation

In order to transition from the introductory and theoretical questions from the previous sections our team implemented multiple scripts to select and gather the raw data from the dataset. To begin answering our discovery questions we had to preprocess our data through a filter the massive dataset by cleaning some administrative tags and filtering out non-English languages. First we performed a keyword search across the "popular_shelves" attribute of the `goodreads_books.json.gz` file. We specifically search for genres: 'romantasy', 'romantic-fantasy', 'fantasy-romance', 'romance', 'fantasy', and 'romantic'. The filtered results are stored in `01_filteredbooks.json`. The resulting json consists of records: book_id, title, average_rating, shelves_list.

To answer discovery question 1, we need to examine the different user interactions, so we implemented a second filtering script utilizing the book_id from our previous script to extract reviews from the `goodreads_reviews_dedup.json.gz` dataset, storing the output in `03_extractedreviews.csv`.

B. Exploratory Data Analysis

Before applying our mining algorithms, our team examined the distributions of ratings, community-defined tags (for example users can create virtual bookshelves and give them a name), and review engagement to establish a baseline understanding of the "Romantasy" and other romance genres subset.

Examining the histogram of the average ratings of romance books in the 01_goodreadsbooks.csv we find that most readers rated the books around 4. This suggests that romance readers find books great reads. They often are not critical of books.

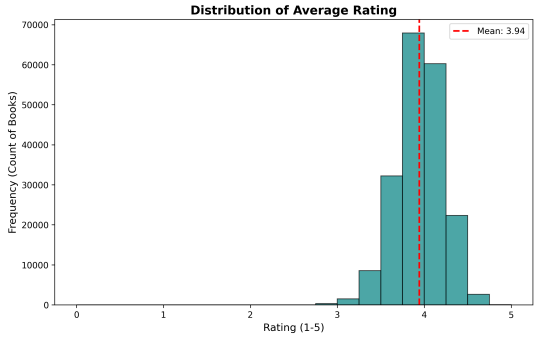

Fig. 2. Distribution of books rating in 01_goodreadsbooks.csv

1) Shelf Tag Distribution: Analysis of the raw `shelves_list` indicated that administrative tags like 'to-read' originally dominated the dataset. During the preprocessing phase, we implemented a filter to remove these generic administrative tags to ensure our algorithms would focus on actual book shelves.

As shown in Figure 3, the most prevalent tags post-cleaning include massive macro-genres such as romance, fiction, and fantasy. Establishing this baseline distribution is critical, as it directly foreshadows our findings for Discovery Question 1, where our association rule mining revealed that the community heavily favors cross-tagging with these broad, overarching categories rather than niche sub-genres.

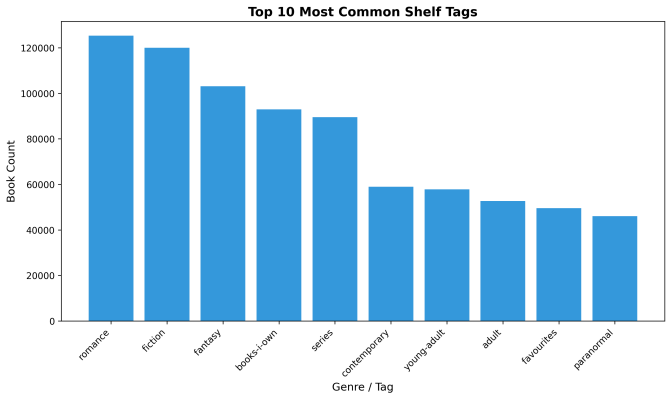

Fig. 3. Distribution of the top 10 most frequent shelf tags post-cleaning

C. Answering Discovery Question 1: Genre Associations

Our team originally proposed if readers of "Romantasy" (Romance Fantasy) also showed high interest in other specific romance sub-genres. During our initial run developing our implementation for Milestone 2 our FP-Growth algorithm generated relationships based off structural administrative tags rather than genre relationships like romance. Furthermore, variations in user-generated text caused our target genre to be heavily diluted. To correct this for Milestone 3, we implemented a data normalization pipeline to aggregate variations of Romantasy (e.g., "fantasy romance", "paranormal romance") into standardized categories. We also significantly lowered our algorithm parameters, dropping the minimum support threshold to 0.01 (1% of transactions) and the minimum lift threshold to 2, to ensure we were not filtering out niche sub-genre connections.

TABLE II
TOP DISCOVERED GENRE ASSOCIATION RULES (MIN SUPPORT = 0.01)

antecedents	consequents	lift	support
supernatural	vampire, paranormal-fantasy	7.520000	0.019000

Even with our highly sensitive thresholds and normalized data, the dataset did not support our initial hypothesis. The FP-Growth algorithm did not find any strong association rules linking 'Romantasy' or other related genres to other specific categories. This suggests that users who tag books with 'Romantasy' tend to treat it as a standalone category. Cross-tagging these books they tend to be associated with broad labels. This null result provides a clear answer to our first discovery question: Readers do not typically engage in sub-genre cross-pollination within the romance category. Instead, "Romantasy" serves as a descriptive tag for these users.

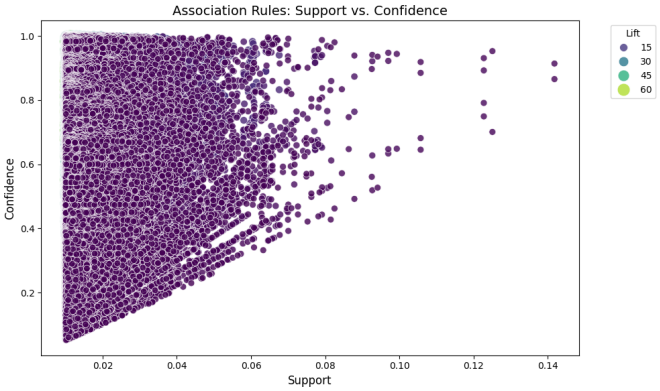

Fig. 4. Genre Associations

VIII. MILESTONE 3

Moving into Milestone 3, our plan remains aligned with the original Milestone 1 proposal. Since we have established our preprocessing pipeline and validated our association rules, our next focus will be applying K-Means clustering to answer our last two discovery questions. We will collect user interactions in an attempt to identify reader personas and find patterns from highly up-voted reviews compared to low rated reviews. However K-means clustering and Pearson's Correlation analysis was only enough to disprove our third discovery question's hypothesis. In order to fully conclude the hypothesis our team had to run the dataset through a LDA model.

A. Discovery Question 2: Reader Personas

The second discovery question explored whether our team could categorize users as "critical reviewers" or "casual consumers" based on average rating variance, review frequency,

and review length. To categorize these users, our team performed a K-Means analysis using Elbow charts to determine the optimal number of clusters based on {review_frequency, avg_rating, rating_variance}.

Based on the Elbow method, we determined that four clusters (Table III) best represent the Goodreads user base: "Generous Consumer," "Harsh Consumer," "Powered Reader," and "Standard Reviewer."

1) Cluster 0: Harsh Consumer: This group is the third largest reading group however this group also seems to read the least. This reading group is categorized by the lowest overall rating of any group. With an average rating of 2.93 and a high variance, it suggests that they either find a rare "5-star" gem or more likely loathe what they read. These are your more difficult to please critics.

2) Cluster 1: Generous Consumer : This is the biggest reading group. Although the readers in this group only rate books occasionally, they seem to love the books they do rate as they have an average rating of 4.44 with a small rating variance of 0.56.

3) Cluster 2: Powered reader: This is the smallest reading group. However despite being the smallest this group averages a review every single day and review the most books out of any group with a review frequency of 321.72. They are fair but tend to be firm in their ratings with a rating variance of 1.08 but only an average rating of 3.78.

4) Cluster 3: Standard reviewer: This is the second biggest reading group. They tend to be more active than the Casual consumers but are more grounded in reviews. This group is pretty standard across the entire board as their average rating of 3.44 shows the books the group reads doesn't leave much of an impact on them. While this group does review a lot of books with a review frequency of 24.65 they tend to rate books uniquely with a rating variance of 1.47.

TABLE III
USER PERSONA PROFILES BY K-MEANS CLUSTER

Cluster	User Count	Review Frequency	Avg. Rating	Rating Variance
0	162270	9.32	2.93	4.62
1	94184	17.01	4.44	0.56
2	4595	321.72	3.78	1.09
3	80856	24.65	3.44	1.47

B. Discovery Question 3: Engagement Volume

Our team's final discovery question asked what patterns emerge from highly up-voted reviews compared to reviews with a lot of down-votes or no votes at all. To answer this, we extracted user engagement metrics from the `goodreads_reviews_dedup.json.gz` dataset to identify behavioral patterns. A Pearson correlation analysis reveals a notable moderate positive correlation of 0.52 between the number of comments (`n_comments`) and the number of votes (`n_votes`), suggesting that reviews sparking discussion are also more likely to be perceived as helpful or popular. Conversely, our initial hypothesis that `review_length` would drive higher engagement was unsupported, as it maintained negligible correlations with both `n_votes` (0.03) and `n_comments` (0.04). Furthermore, the numerical `rating` appears entirely decoupled from interaction volume, showing correlations of 0.00 and -0.01 against votes and comments, respectively. These results indicate that while social interactions (voting and commenting) tend to scale together, they are driven by factors other than the sentiment of the rating or the verbosity of the review.

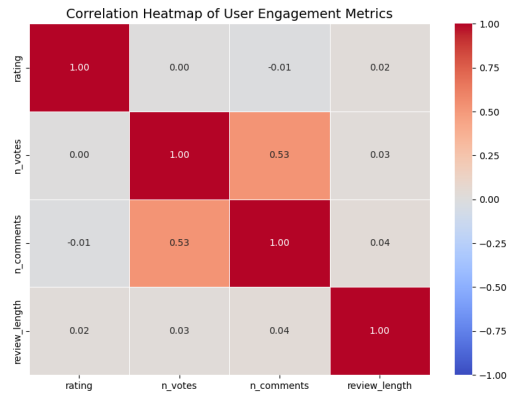

Fig. 5. Correlation Heatmap of User Engagement Metrics

Visualizing this relationship via a scatterplot (Figure 6) confirms the heatmap findings. While the majority of reviews cluster near the origin of the graph, there is a clear positive trend where reviews exceeding $1,000$ votes frequently receive over 150 comments. These results indicate that while social interactions (voting and commenting) tend to scale together, they are driven by the community relevance of a review rather than the sentiment of the rating or the content of a certain book.

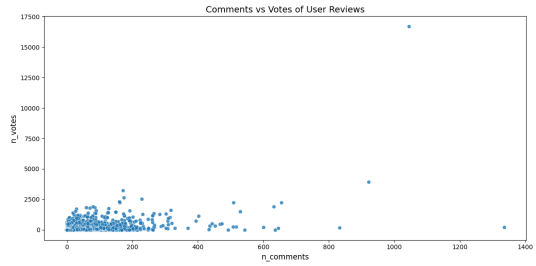

Fig. 6. Scatterplot of Comments vs. Votes

1) Latent Dirichlet Allocation (LDA) : Because our Milestone 3 findings proved that longer reviews and star ratings do not correlate with higher community engagement, our team decided to transition to qualitative text mining to answer our final discovery question. We utilized Latent Dirichlet Allocation (LDA), This is a machine learning algorithm designed to tokenize a massive dataset to find patterns and divide them up.

Our team first had to prepare our review dataset. Importing our `03_extractedreviews.csv` into a Pandas dataframe, we tokenized the review text while simultaneously

stripping non-alphabetic characters and filtering out custom domain stop-words (e.g., 'book', 'read', 'story') that add no value. After converting our cleaned tokens into a matrix, we built an LDA model. To ensure we extracted concise and highly specific themes, we configured the model to search for 40 distinct components ($n_components = 40$). We then mapped these generated topics against our engagement metrics (`n_votes`, `n_comments`, and `review_count`).

Reviewing our modeled data allows us to identify several distinct themes that drive community interaction. While our team identified 40 topics a few of the big themes were Analytical critique, Emotional Reactionaires and Administrative / ARC disclaimers.

- **Analytical critique (Topics 7, 13 % 22)** These topics mention specifically world building and pacing of the books. These books averaged about 0.22-1.93 votes and 0.29-0.52 comments. This suggests that the analytical critique reviews are popular with the goodreads community and are favored greatly.
- **Emotional Reactionaries (Topics 18 & 34):** These topics captured reviews focused entirely on feelings, specifically mentioning "book hangovers" and crying. Essentially, the readers who wrote these reviews loved the books so much they could not seem to get over the ending. Interestingly, these reviews saw notably lower engagement than the analytical critiques, averaging between 0.79 and 0.80 votes, and 0.18 to 0.37 comments.
- **Administrative / ARC disclaimers (Topic 35):** Topic 35 was an interesting one. Despite our team doing our best to pre-process our datasets and remove administrative tags a few reviews were prevalent enough to be included. These readers talked about their blog or disclosing advanced reader's copies before reviewing. Most authors hand out Advanced Reader's Copies (ARC) for them to get a bunch of reviews before a product launch. Since the expectation for readers with ARCs is to review books our team expects to have high levels of engagement. Furthermore, this group had a `n_votes` value of 1.76 while `n_comments` had a value of 0.64.

IX. Milestone 4

Milestone 4 will focus on refinement and re-factoring code. We plan to present our finished project for milestone 4 so our team will focus on cleaning up the report and the Juptyer notebooks to get a finished final product. We plan to look at all the feedback from the other three milestones to come up with a finished product.

Acknowledgment

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References

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